Tiancheng LIU

Ph.D. Candidate at The Hong Kong University of Science and Technology (Guangzhou)
Visiting Researcher at University of Amsterdam
tcliu.cc | tcliu767@connect.hkust-gz.edu.cn | +86 1521-8828-949

ABOUT ME

Tiancheng Liu is a computational media researcher and artist working across Creative AI, multimodal interaction, and culturally grounded computing. His work investigates how machines can understand, sense, and participate in human creative practices, with a particular focus on embodied cultural forms such as Chinese calligraphy. His Ph.D. research is organized around three interrelated layers of creative practice. The first layer asks how machines can analyze and interpret the aesthetic structure of finished artifacts, including layout, form, balance, and stylistic preference. The second layer asks how machines can sense the often invisible processes behind creation, such as gesture, pressure, rhythm, and other implicit micro-actions. The third layer asks how AI systems can join creative practice as collaborators, supporting co-creation, performance, and expressive interaction. Building upon this framework, he seeks to explore broader possibilities for computing approaches as a medium for interpretation, expression, and collaboration, expanding how humans and machines can sense, make, and imagine together in the cultural and artistic context.

EDUCATION

University of Amsterdam, Amsterdam, Netherlands — 04.2026 – Present
Visiting Researcher in MultiX, Informatics Institute

HKUST (GZ), Guangzhou / Hong Kong SAR, China — 09.2022 – Present
Ph.D. Candidate in Computational Media and Arts, Information Hub

Waseda University, Tokyo, Japan — 09.2018 – 10.2020
M.Eng in Information

South China University of Technology, Guangzhou, China — 09.2014 – 07.2018
B.Eng in Network Engineering

South China University of Technology, Guangzhou, China — 02.2015 – 07.2018
B.Econ in Finance

Zhixin High School Guangzhou, China — 09.2011 – 06.2014

PUBLICATIONS

Accepted

[1] **Tiancheng Liu**, Nanne Van Noord, Chen Liang, and Kang Zhang. "Full-Page Chinese Calligraphy Character Detection: A Dataset and Benchmark with Integrity-Preserving Tiling and SDAC-Det". In: *VISART VIII: Vision for Art and Culture Workshop, European Conference on Computer Vision (ECCV)*. Accepted; DOI TBD. Malmö, Sweden, 2026.

[2] **Tiancheng Liu**, Shumeng Zhang, Hanyu Qu, and Hanqin Zhou. "NarraStone: Reimagining the Stone Stele as an AI-Era Apparatus of Synthetic Memory". In: *Proceedings of the SIGGRAPH Asia 2026 Art Papers*. SA Art Papers '26. Accepted; DOI TBD. Kuala Lumpur, Malaysia: Association for Computing Machinery, 2026.

[3] Wenjing Mao, Han Lin, **Tiancheng Liu**, and Lujie Zheng. "Future Sample: A Participatory Installation for Encountering AI-Generated Alternative Selves". In: *VINCI '26: The*

19th International Symposium on Visual Information Communication and Interaction, Art Gallery Track. Accepted; DOI TBD. Hangzhou, China, 2026.

Published

[1] **Tiancheng Liu**, Shumeng Zhang, and Nicolò Merendino. "BruSHŪ: Cross-Modal Translation of Implicit Micro-Actions in Chinese Calligraphy". In: *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 9.3 (2026). To be presented at SIGGRAPH 2026, Los Angeles, CA, 19–23 July 2026, pp. 1–13. [Online].

[2] Piera Riccio, Selina Khan, Ludovica Schaerf, Shuai Wang, **Tiancheng Liu**, Athanasios Efthymiou, Noa Garcia, and Nanne van Noord. "Understanding Art & Culture". In: *Proceedings of the 2026 International Conference on Multimedia Retrieval*. ICMR '26. Association for Computing Machinery, 2026, pp. 2901–2903. [Online].

[3] Shumeng Zhang, Shiqi Lin, **Tiancheng Liu**, Mingming Fan, and Raul Masu. "To Perform/To Live: Decolonizing Digital Music Instruments and Feminism with Human-AI Co-Created Embodied Experience of Daily Objects". In: *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 9.3 (2026). To be presented at SIGGRAPH 2026, Los Angeles, CA, 19–23 July 2026, pp. 1–12. [Online].

[4] **Tiancheng Liu**, Siyu Yan, Shumeng Zhang, Chen Liang, and Kang Zhang. "Computational Interpretation of Chinese Calligraphy Layout via Graph-Based Modeling and Language Models". In: *Proceedings of the POLYU COMP - HKUST(GZ) INFH Research Student Conference*. Poster presentation. Hong Kong, China, 2025.

[5] **Tiancheng Liu**, Jiayi Ye, Shumeng Zhang, Kang Zhang, and Chen Liang. "Quantifying Structural Aesthetic Features and Personality Trait Preferences in Kai Shu Calligraphy". In: *Proceedings of the 33rd ACM International Conference on Multimedia*. MM '25. Dublin, Ireland: Association for Computing Machinery, 2025, pp. 6730–6739. [Online].

[6] Yinzong Wei, Yiming Fu, Siqi Wang, Shuwei Gao, **Tiancheng Liu**, Zeyu Wang, and Xin Tong. "A Deep-Learning Model for Edition Identification of Premodern Chinese Rare Books". In: *Digital Humanities* 03 (2025). [Online], pp. 1–10.

[7] Shumeng Zhang, **Tiancheng Liu**, and Mingming Fan. "FinHERtip: Embodied Identity and Human-AI Co-Creation in Accessible Musical Performance". In: *Proceedings of the SIGGRAPH Asia 2025 Art Papers*. SA Art Papers '25. Association for Computing Machinery, 2025. [Online].

[8] **Tiancheng Liu**, Ling Li, and Manling Yang. "Chinese Seal Carving Aesthetic Evaluation". In: *Proceedings of the 17th International Symposium on Visual Information Communication and Interaction*. VINCI '24. Association for Computing Machinery, 2024. [Online].

[9] **Tiancheng Liu**, Anqi Wang, Xinda Chen, Jing Yan, Yin Li, Pan Hui, and Kang Zhang. "PoEmotion: Can AI Utilize Chinese Calligraphy to Express Emotion from Poems?" In: *Proceedings of the 29th International Symposium on Electronic Arts: ISEA2024 Everywhen Proceedings. Volume 1: Academic Papers*. Ed. by Gavin Sade, Andrew Brown, Leah Barclay, Jen Seevinck, Anastasia Tyurina, and Rewa Wright. Brisbane, Qld: Queensland University of Technology and ISEA International, 2024, pp. 476–483. [Online].

[10] Aven Le Zhou, Jiayi Ye, **Tiancheng Liu**, and Kang Zhang. "Archiving Body Movements: Collective Generation of Chinese Calligraphy". In: *Proceedings of the 29th International Symposium on Electronic Arts: ISEA2024 Everywhen Proceedings. Volume 1: Academic Papers*. Ed. by Gavin Sade, Andrew Brown, Leah Barclay, Jen Seevinck, Anastasia Tyurina, and Rewa Wright. Brisbane, Qld: Queensland University of Technology and ISEA International, 2024, pp. 862–870. [Online].

[11] **Tiancheng Liu** and Keiichi Koyanagi. "Blockchain Technology Application in Access Control". In: *Proceedings of the 6th Waseda ORIS International Seminar for Junior Researchers*. Waseda University. Tokyo, Japan, 2020.

[12] Helong Li, Tianqi Luo, Liuling Li, and **Tiancheng Liu**. "Measuring Systemic Risk in the Chinese Financial System Based on Asymmetric Exponential Power Distribution". In: *Recent Developments in Data Science and Business Analytics*. Ed. by Madjid Tavana and Srikanta Patnaik. Cham: Springer International Publishing, 2018, pp. 225–232. [Online].

EXHIBITIONS

Ars Electronica 2026, presenting *BruSHŪ*, Linz, Austria, 09.2026
SYMBOTIC SIGNALS, presenting *BruSHŪ*, Guangzhou, China 06.2026
InnoCharity, presenting *BruSHŪ*, Guangzhou, China 05.2026

GRANTS AND FUNDING SUPPORT

Project Researcher (Proposal Co-author) — National Social Science Fund of China (NSSFC) Major Project
HKUST (GZ) 2024–present
Topic: Application of Frontier Technology in Cultural Heritage Protection and Inheritance

- **Program:** Special Program on Cultural Heritage Protection and Inheritance Research
- **Co-applicants:** PI: Prof. Kang ZHANG
- **Grant:** No. 24VWB020

Recipient — Overseas Research Travel Grant (RTG)
HKUST (GZ) 2024, 2025 & 2026
Topic: Travel to present research at international conferences (outside Greater China)

- **Role:** Recipient
- **Funding:** RTG conference travel support

Recipient — Overseas Study Award
HKUST (GZ) 2026
Topic: Visiting researcher appointment at University of Amsterdam (UvA)

- **Role:** Recipient
- **Funding:** Overseas study support

AWARDS & SCHOLARSHIP

First Prize, Model Creativity Competition, by *Red Bird Maker Space, College of Future Technology, Hong Kong University of Science and Technology (Guangzhou)* 2024
Finalist, Three Minute Thesis (3MT) Competition, by *Hong Kong University of Science and Technology (Guangzhou)*, founded by The University of Queensland 2024
Postgraduate Studentship (PGS), research studentship awarded by *Hong Kong University of Science and Technology (Guangzhou)* 2022 – Present
JASSO Scholarship — *School of Information, Production and System, Waseda University*,

Tokyo, Japan 2018 – 2020
Second-Class Academic Scholarship — *School of Computer Science and Engineering, South China University of Technology*, Guangzhou, China 2015 – 2017

ACADEMIC SERVICE

- **Tutorial Contributor and Presenter**, "Understanding Art & Culture," ACM International Conference on Multimedia Retrieval (ICMR), Amsterdam, Netherlands, 2026. Contributed and presented a segment on Chinese calligraphy as a case for multimodal art understanding, covering aesthetic structure, implicit micro-actions, and human-AI co-creation.
- **Web Chair** of The 16th International Symposium on Visual Information Communication and Interaction, VINCI 2023, Guangzhou, China
- **Serving as reviewer for journals and conferences** in visual computing/graphics & imaging, information visualization and HCI, AI and society, and electronic & information engineering:
  - CONF.: European Conference on Computer Vision (ECCV); ACM International Conference on Multimedia (ACM MM); International Symposium on Visual Information Communication and Interaction (VINCI);
  - JOURN.: International Journal of Human–Computer Interaction (IJHCI); The Visual Computer; AI & Society; Frontiers of Information Technology & Electronic Engineering; Visual Computing for Industry, Biomedicine, and Art (VCIBA).

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant / Guest Instructor *HKUST(GZ)*

- *Introduction to Experimental Animation* 2023
  - Assisted in course design, delivered lectures;
  - Guided students in stop-motion animation shooting, simple animation producing, video editing and creative coding in visual computing projects.
- *Interactive Storytelling* 2024
  - Supervised student projects combining narrative design, human–computer interaction, and generative AI techniques;
  - Provided hands-on training in relevant software tools and digital media techniques, improving students' production skills.

Research Mentor in MPhil's Program, *HKUST(GZ)*

- *Research and Artistic Creation on the Aesthetics and Dissemination of Chinese Calligraphy and Scripts* 2023–2025.
  - Supervised postgraduate research on computational interpretation and aesthetic modeling of Chinese calligraphy;
  - Guided interdisciplinary collaboration between visual computing and digital humanities.
- *Multi-Modal Embodied Human–Pet Interaction via Pet Wearable Devices* 2025–present
  - Managing an interdisciplinary research project integrating embodied interaction design, multi-modal data analysis, and AI-driven affective computing for human–pet communication.

INDUSTRY R&D EXPERIENCE

Network R&D Center, China Telecom Research Institute Beijing/Guangzhou, China
Applied R&D Engineer 2021 – 2022
- Built an LSTM-based traffic-flow forecasting model for short-term prediction from real-time

sensor data.
- Developed a reinforcement-learning model for signal-timing optimization to improve traffic-light control at intersections.
- Trained license-plate detection and recognition models on in-house data, using YOLO for detection and an OCR model for plate reading.
- Helped develop the MQTT data pipeline, including publish/subscribe workflows and connection handling, to feed models with real-time edge and roadside data; also contributed to front-end product design and requirements breakdown with product and platform teams.

Cloud Network Institute, China Telecom Research Institute Guangzhou, China
Cloud Network Researcher 2020 – 2021
- Drafted an initial ITU-T testing and monitoring framework for cloud-network architecture, translating requirements into test methods, test cases, metrics, and a standard reporting format through internal review and revision.
- Built a Prometheus-based prototype to run the framework in practice, supporting batch test execution and longitudinal key-metric collection.

GF Securities Guangzhou, China
Quantitative Researcher 2018
- Researched low-risk asset allocation for fund management using regression analysis and linear optimization, translating risk and compliance limits into a workable strategy.
- Built a prototype backtesting and risk-control system covering data processing, machine-learning factor extraction including LSTM, and strategy execution against historical data.

PUBLIC ENGAGEMENT, OUTREACH & LEADERSHIP

STEM outreach Lecturer — *3-Day AI Workshop for Teenagers*
Guangzhou, China *2026*
- Topic: how AI understand world — hands-on demos linking machine perception with creative practice;
- Designed and delivered a 3-day lecture series introducing AI concepts to high school students.

Invited Speaker — Salon: "*The Aesthetics of Innovation and Entrepreneurship in Nansha*"
Guangzhou, China *2025*
- Topic: "Seeing–Feeling–Gaining" perspectives on Nansha's innovation and entrepreneurship ecosystem.

Invited Speaker — "*'Qing Chu Yu Nan' Talent Gathering Action Plan*" Launch Event
Guangzhou, China *2025*
- Topic: Youth-friendly innovation ecosystem in Nansha; tech integration (e.g., policy support for researchers).

STEM Outreach Lecturer — Tongren No. 1 High School
Tongren, Guizhou, China *2023*
- Designed and delivered two short courses: (1) Artificial Intelligence Technology and Calligraphy Appreciation; (2) Sustainable Green Building and Future Urban Development.
- Led hands-on activities (e.g., AI demo for calligraphy style recognition) and facilitated a student showcase.

Founder — Campus Fitness Community, HKUST (Guangzhou)
Guangzhou, China *2022–Present*
- Founded and led a 300 + member student-staff community; organized group workouts and wellness events.

- Coordinated scheduling and safety protocols with the university gym; supported facility operations.
- Promoted campus wellness initiatives and peer coaching.

Solution Presenter — *ODCC Edge Computing Conference 2021*
Beijing, China *2021*

- Talk: "China Telecom Cloud–Network Convergence Innovation Platform: R&D and Application Practice."

Director — Student Art Union, South China University of Technology
Guangzhou, China *2017–2018*

- Organized and executed campus performances and events;
- Coordinated cross-department collaborations;
- Supervised teams' daily training and budgeting.

LANGUAGES

Mandarin, English, Cantonese, Japanese (N2 Level)

HOBBIES

Bamboo Flute, Calligraphy, Trekking, Triathlon, Museum, Urban Exploration